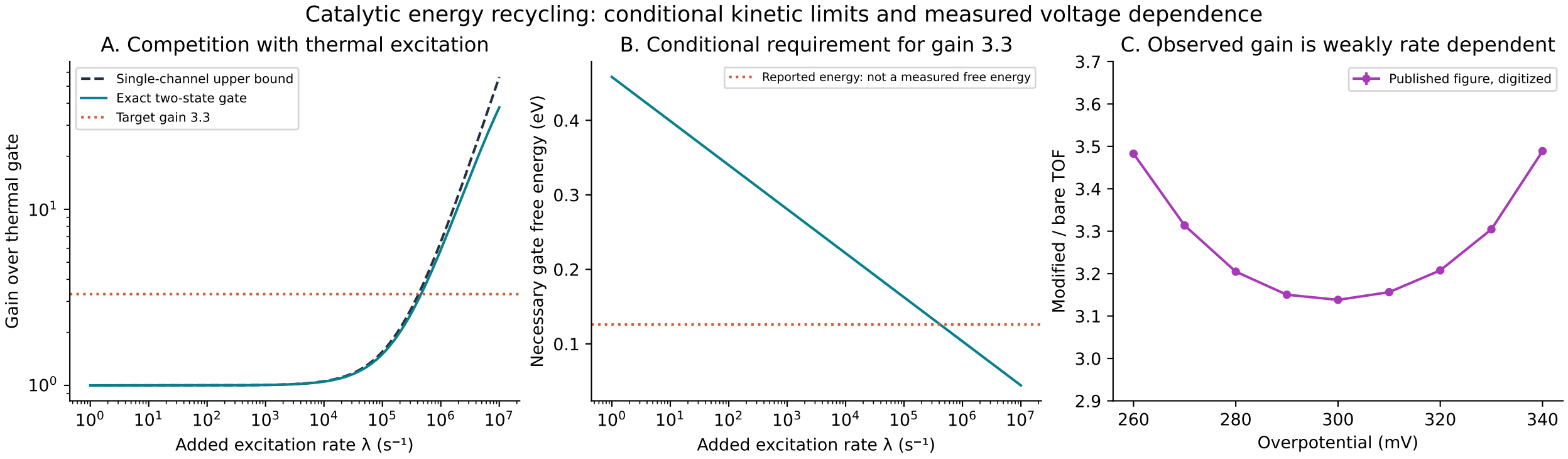
Catalytic energy recycling: conditional kinetic limits and measured voltage dependence
A. Competition with thermal excitation
Single-channel upper bound
Exact two-state gate
Target gain 3.3
Gain over thermal gate
Added excitation rate λ (s⁻¹)
B. Conditional requirement for gain 3.3
Reported energy: not a measured free energy
Necessary gate free energy (eV)
Added excitation rate λ (s⁻¹)
C. Observed gain is weakly rate dependent
Published figure, digitized
Modified / bare TOF
Overpotential (mV)